

metaDEBASS — who do you trust tonight?

Trust-aware fusion of transient classifiers for Rubin/LSST follow-up



Problem: early follow-up is a trust problem

- LSST will deliver $\sim 10\text{M}$ alerts/night scored by many broker classifiers that often disagree.
- Follow-up observers need to know **which classifier to trust on which object**.
- The decision has to happen after only **3–5 detections**.

Primary product

Per-expert confidence at the current object epoch.

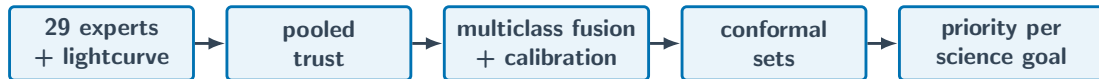
Secondary output

A trust-weighted follow-up proxy for tonight's science goal.

Truth boundary

Broker-derived labels stay weak; final science truth stays separate.

Architecture: confidence first, priority second



Expert confidence

Learn $P(\text{expert right} \mid \text{object, epoch})$ before asking for a fused class.

Goal conditioning

Rank with $u \cdot p$ after calibration and abstain when conformal sets stay broad.

Honesty machinery

- Pre-registered headline and guards.
- Locked test split byte-identical across versions.
- Component gates decided on calibration with bootstrap CIs.
- Label-provenance anti-circularity: weak labels are not science truth.
- Independent audits: adversarial re-computation and code audit.

Headline protocol: object-level, spectroscopic-only, locked test split, $n_{\text{det}}=5$, 1000-resample paired object bootstrap.

Label provenance bug fixed

3,149

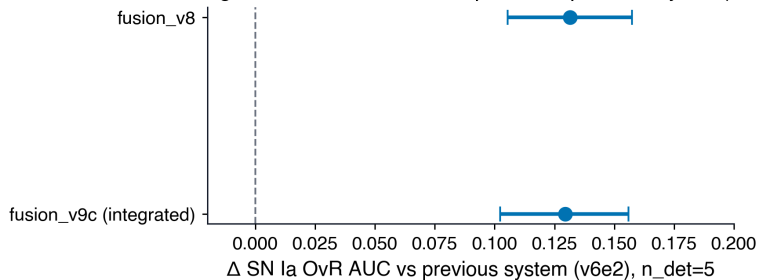
BTS-unclassified objects carried fabricated nonla/spectroscopic labels.

15

were in the locked test; all were demoted.

Headline: pre-registered gain over the previous system

Pre-registered headline — 765 spectroscopic test objects (SCC)



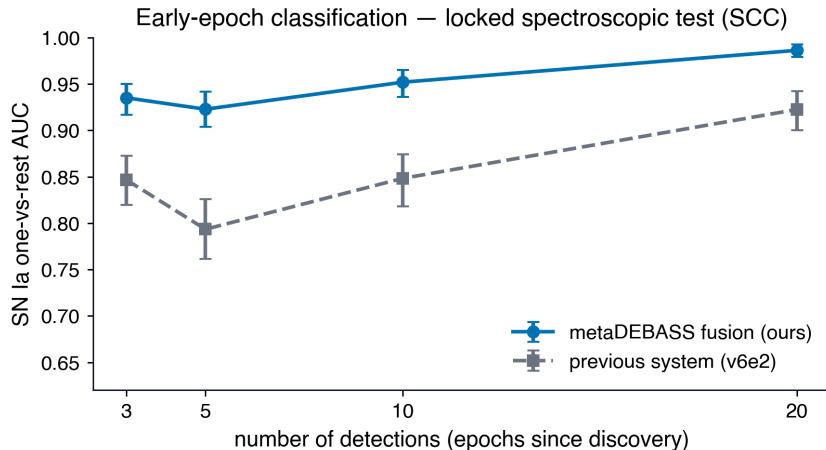
Delta SN Ia OvR AUC
+0.129 [0.102, 0.156]

Spectroscopic test objects
765

Object-level, spectroscopic-only, locked test, n_det=5.

Macro AUC
0.921

AUC vs n_det: early-epoch gains persist



At **n_det=5**

SN Ia OvR AUC is
0.923 vs 0.793 for the
previous system.

At **n_det=20**

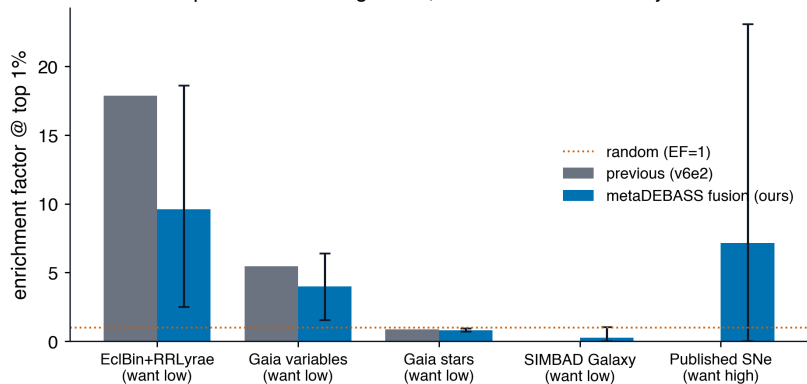
SN Ia OvR AUC is
0.986 vs 0.923.

Epoch grid

n_det values: 3, 5, 10,
20.

DP1 enrichment: rank for science, expose contaminants

Operational ranking on 15,868 real Rubin DP1 objects



EclBin+RRLyrae
17.9 → 9.6

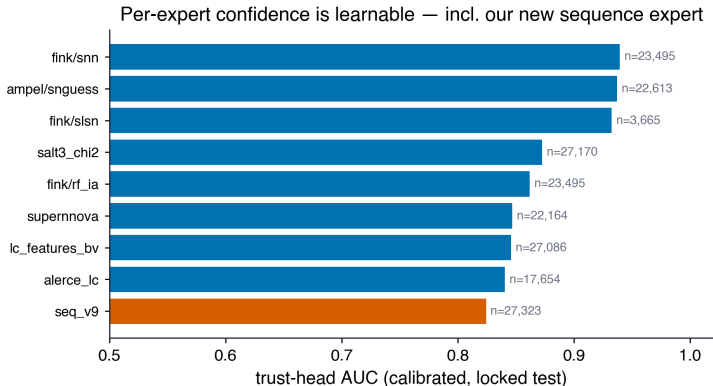
Gaia variables
5.5 → 4.0

Published SNe
0.0 → 7.1

EF=1 is random; contaminant classes want low, Published SNe wants high.

N=15,868 DP1 objects.

Trust heads: per-expert confidence is learnable



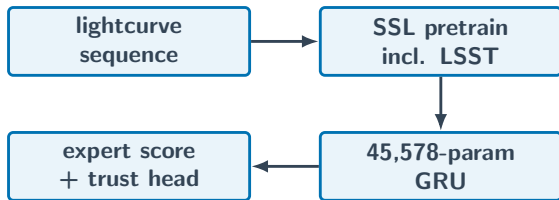
seq_v9 highlighted

Trust-head AUC 0.824 on 27,323
locked-test rows.

Coverage

13 trust-headed experts out of 29
registered experts.

The v9 sequence expert



Standalone AUC at $n_{\text{det}}=5$
0.856 / 0.865

Trust-head coverage
27,323 test rows

Role in fusion

Useful as a calibrated trust signal; fusion gate still reports its standalone component result.

Gate ledger: we report what failed

Component	Decision	Delta macro AUC [CI95]	Readout
seq_features	skipped	not scored	not promoted
ext_features	kept	+0.0087 [0.0021, 0.0160]	clear cal gain
seq_v9_expert	dropped	-0.0009 [-0.0049, 0.0031]	no cal lift
traj_features	dropped	+0.0005 [-0.0036, 0.0049]	CI crosses zero
pooled_trust_q	dropped	-0.0569 [-0.0734, -0.0391]	harmful
expert_dropout_aug	dropped	-0.0025 [-0.0057, 0.0011]	no cal lift
dirichlet_calibration	kept	no delta	calibration guardrail

Deliverables and next steps

Built this week

Gold snapshots: 211,392 rows across 12,772 objects.

Broker layer

29 experts registered; 13 have trust heads.

Data scale

8,774 ZTF objects plus 3,998 LSST objects.

Compute

v9c rerun: 54 min after v8 at 6.0 SCC hours.

- Broaden broker history beyond the ALeRCE-heavy seed set.
- Make epoch-safe broker snapshots the default trust-training input.
- Convert conformal output into observing policies for different science goals.
- Promote sequence components only when calibration gates show positive lift.